

certified power level.

Thermal effects. Tissue heating is the principal mechanism of interaction between radio-frequency energy and the human body. Tissue is a dielectric material which is heated by rotations of polar molecules induced by the microwave radiation.

At the frequencies used by mobile phones, most of the energy is absorbed by the skin and other superficial tissues, resulting in negligible temperature rise in the brain or any other organs of the body. In the case of a person using a cellphone, most of the heating effect occurs at the surface of the head, causing its temperature to increase by a fraction of a degree. The local blood flow mechanism of the brain is capable of disposing of excess heat. However, the cornea of eye does not have temperature regulation mechanism and is more prone to produce cataracts.

Long-duration cellphone usage also causes burning effects to the ear and earaches, or sharp pains inside the ear to the same side of the head as the phone is held. Some users even experience pain descending down from just behind the ear onto the jaw. Some users feel numbing sensations or tingling mainly in the outer area of ear, face or jaw.

Eye problems may include fluttering of the eyelids, blurring of vision and bloodshot eye—always in the eye nearest the cellphone. Several studies have indicated that low-level RF exposure due to cellphones could cause significant visual effects, including the destruction of corneal endothelial cells, increased vascular permeability and destruction of photoreceptors in the retina.

Non-thermal effects. Some biophysicists argue that there are several thermoreceptor molecules in cells, which activate a cascade of second and third messenger systems, gene expression mechanisms and production of heat-shock proteins in order to defend the cells against metabolic cell stress caused by heat. The increases in temperature that cause these changes are too small to be detected.

Some studies indicated that stress proteins are unrelated to thermal ef-

fects. These studies proved that stress proteins occur for extremely low frequencies (ELF) and radio frequencies which have very different energy levels. Some studies indicated a leakage of albumin into the brain via a permeated blood-brain barrier. Several other studies indicated that the blood-brain barrier breaks down with exposure to low-intensity cellphone frequency.

Cognitive effects. The impact of cellphone radiation on cognitive functions of humans has been studied by several researchers. These studies confirmed slower response to a spatial working memory task when exposed to cellphone radiation.

The Defence Evaluation and Research Agency (DERA), the UK's largest science and technology organisation, has conducted research on cellphones' EM radiation on the brain cells of rats and showed loss of memory for short periods. Several users of mobile phones also reported short-term memory loss. The cognitive functions are heavily affected by longer exposure.

EM allergic reaction. A number of studies have investigated the effects of mobile phone radiation specifically during and after its use. The users reported several unspecific symptoms like burning and tingling sensations in the skin of the head and extremities, sleep disturbances, dizziness, fatigue, loss of mental attention, reaction times and memory retentiveness, headaches, heart palpitations, blood pressure and disturbances in digestive system.

Most users reported pain or ache on the same side as they held the phone and normally in the temporal region. When pain is located on one side of the head, users can invariably reproduce it on the other side by switching ears in an effort to obtain temporary relief.

Genotoxic effects. Cellphone RF radiation at both 900 MHz and 1800 MHz and SAR of 0.4-27.5W/kg showed a correlation between increasing SAR and decreased motility and vitality in sperm, increased oxidative stress, stimulating DNA base adduct formation and increased DNA fragmentation. A European study named

REFLEX (Risk Evaluation of Potential Environmental Hazards from Low Energy Electromagnetic Field Exposure Using Sensitive in-vitro Methods) indicated some evidence of DNA damage of cells in in-vitro cultures, damage to chromosomes, boosted rate of cell division and alterations in the activity of certain genes.

Cancer. A multinational case-control study on adults, INTERPHONE, coordinated by the International Agency for Research on Cancer (IARC), investigated whether the RF fields emitted by cellphones increased the risk of cancer. The study focused on tumours arising in the tissues most exposed to RF fields from mobile phones: glioma and meningioma (the most common types of brain tumours), acoustic neurinoma and parotid gland tumours. It found no increased risk of glioma or meningioma but there were some indications of an increased risk of glioma for those who reported the highest 10 per cent of cumulative hours of cellphone use.

However, it has been suggested that EM fields associated with cellphones may play a role in speeding up the development of an existing cancer. Some studies reported lump in the neck or (lymphomas) also.

Recommendations

To date, research does not suggest any consistent evidence of adverse health effects from exposure to EM fields at levels below those that cause tissue heating. The WHO has issued a precautionary principle applied in circumstances with a high degree of scientific uncertainty, reflecting the need to take action for a potentially serious risk without awaiting the results of scientific research. Some recommendations to minimise the possible health hazards are:

1. Keep the cellphone as far away from your body as possible. By moving the cellphone just 5 cm away from your head while talking on it, electromagnetic radiation that reaches to head is reduced by 75 per cent.
2. Turn your cellphone off at night because EM radiation emitted by the cellphone can disturb sleep quality.